Dense critical graphs from saturated graphs

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k-critical graph on n vertices. For every fixed integer $k \geq 8$, we construct a sequence of k-critical graphs G_s with $|V(G_s)| \to \infty$ and

$$\frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \begin{cases} \dfrac{k-4}{2(k-2)}, & k \text{ even}, \\ \dfrac{k-5}{2(k-3)}, & k \text{ odd}. \end{cases}$$

For every $k \geq 8$ except $k = 9$, this improves the best known lower bounds, due to Toft (1970). For k divisible by three and at least twelve, it disproves the asymptotic formula proposed in Erdős Problem 917, previously known to fail when $3 \nmid k$. The smallest new case for the formula is $k = 12$, with density $2/5 > 3/8$. The construction combines a coloring module of Pegden with saturated graphs of sublinear maximum degree. The complete proof is formalized and verified in Lean 4; generative AI was used in developing the arguments and the manuscript.

1 Introduction

All graphs are finite and simple. A graph is k-critical if it has chromatic number k and every proper subgraph is $(k-1)$-colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the maximum number of edges of a k-critical graph on n vertices (in the edge-deletion sense). Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2}\left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2. \tag{1.1}$$

This is the general asymptotic question in Erdős Problem 917 [4]. Dirac [2, p. 90] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős [3, p. 28, eq. (3)] observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [8] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1, pp. 97–98] and Jensen [5]. Luo, Ma, and Yang [7] noted that no constructions improving Toft's asymptotic constants had been found since 1970; their Theorem 1.1 gives upper bounds on $f_k(n)$. We give the following construction.

Theorem 1. *For every integer $k \geq 8$, there is a sequence of k-critical graphs G_s such that $|V(G_s)| \to \infty$ and*

$$\frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \begin{cases} \dfrac{k-4}{2(k-2)}, & k \text{ even}, \\ \dfrac{k-5}{2(k-3)}, & k \text{ odd}. \end{cases}$$

Write c_k for the density in Theorem 1. Toft's coefficients [8], in the notation of [7, Section 1], are

$$b_k = \frac{1}{2} - \frac{3}{2k - \delta_k}, \qquad \delta_k = \begin{cases} 0, & k \equiv 0 \pmod 3, \\ 8/7, & k \equiv 1 \pmod 3, \\ 44/23, & k \equiv 2 \pmod 3. \end{cases}$$

Corollary 2. *For every $k \geq 8$, $k \neq 9$, the density c_k exceeds Toft's coefficient b_k. Moreover, for $3 \mid k$ and $k \geq 12$, (1.1) fails. For $k = 9$, all three coefficients equal $1/3$.*

Proof. Theorem 1 and $f_k(|V(G_s)|) \geq e(G_s)$ give $\limsup_{n\to\infty} f_k(n)/n^2 \geq c_k$. For even $k \geq 8$, the inequality $c_k > b_k$ is equivalent to $k > 6 - \delta_k$; for odd $k \geq 11$, it is equivalent to $k > 9 - \delta_k$. Both inequalities hold. When $3 \mid k$, Toft's coefficient is $b_k = \frac{1}{2}(1 - 1/\lfloor k/3 \rfloor)$. Formula (1.1) would force the normalized limit, and hence the limsup, to equal b_k, contradicting the lower bound for $k \geq 12$. At $k = 9$, all three coefficients equal $1/3$. □

For even $k = 2t + 2$, we combine t copies of Pegden's module [9], joining their active sets by an almost complete t-partite graph. The omitted edges are prescribed by a K_t-saturated graph H of small maximum degree [1]. In a $(2t+1)$-coloring, each active set needs three colors and the degree bound forces two to be private to that set. The remaining common color would give a K_t in H, a contradiction. Saturation supplies the colorings after an edge between modules is deleted. Odd k follow by joining with K_1.

2 A coloring module

Let $r \geq 3$, let S be r-critical, and let T be $(r-1)$-critical. The module $U(S,T)$ consists of disjoint copies of S and T, together with an independent set

$$A = V(S) \times V(T).$$

Each $(x, y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T, and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x.

The following is the $U(r, r-1)$ case of Pegden's module lemma [9, Lemma 2.5], stated there for triangle-free factors; the lemma holds for general critical factors S and T of the indicated chromatic numbers.

Lemma 3. *Fix a palette of r colors.*

1. *In every proper coloring of $U(S,T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ, there is a proper coloring in which the active set uses only α, β, γ, and γ occurs on the active set only at a_0.*
3. *Given an edge e of $U(S,T)$ and distinct colors α, β, there is a proper coloring of $U(S,T) - e$ in which the active set uses only α, β.*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all r colors, it contains a vertex colored α and a vertex colored β. Every active vertex in the first row must have color β, so no vertex of T can have color β. The second row similarly excludes α from T. This leaves only $r - 2$ colors for T, a contradiction.

For (2), color S so that α occurs only at x_0. Color T with the $r - 1$ colors other than α, so that β occurs only at y_0. These colorings exist by criticality: delete the specified vertex, color the remaining graph with one fewer color, and give the deleted vertex its own color. Extend these colorings by

$$\operatorname{col}(x,y)=\begin{cases}\alpha, & x\neq x_0,\\ \beta, & x=x_0,\ y\neq y_0,\\ \gamma, & (x,y)=(x_0,y_0).\end{cases}\tag{2.1}$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e\in E(S)$, color both $S-e$ and T with the $r-1$ colors other than α, and color all of A with α. If $e\in E(T)$, color $T-e$ with the $r-2$ colors other than α,β. Choose $x_0\in V(S)$ and color S with α only at x_0. Color the row indexed by x_0 with β and all other rows with α.

Suppose instead that e joins an active vertex (x_0,y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0,y_0) to x_0, use (2.1) with the color of (x_0,y_0) changed to α. If it joins (x_0,y_0) to y_0, change that color to β. In either case the only edge that would become monochromatic is the deleted edge. □

3 From saturated graphs to critical graphs

A graph is K_t-saturated if it contains no K_t, but adding any missing edge creates a K_t. Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a K_{t-2}.

Proposition 4. *Let $t\geq 3$, and let H be a K_t-saturated graph on $v\geq t$ vertices with maximum degree $d<v-1$. Let $h\geq 2t+1$ be odd and suppose*

$$v>2t(t-1)hd.\tag{3.1}$$

Put $a=2hv$. There is a $(2t+2)$-critical graph G with

$$|V(G)|=t(a+h+2v)\tag{3.2}$$

and

$$e(G)\geq\binom{t}{2}a^2\left(1-\frac{d}{v}\right).\tag{3.3}$$

Proof. Construction. Set $\ell=2v$ and take joins of cliques and odd cycles, as in [2, p. 90]:

$$S=K_{2t-2}\vee C_{h-2t+2},\qquad T=K_{2t-3}\vee C_{\ell-2t+3}.$$

Both cycles are odd and have length at least three. Since the join of a p-critical and a q-critical graph is $(p+q)$-critical, S is $(2t+1)$-critical and T is $2t$-critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take t disjoint copies $U_1,\dots,U_t$ of $U(S,T)$, with active sets $A_1,\dots,A_t$, each of size $a=h\ell$. In each copy, identify $V(T)$ with $V(H)\times\{0,1\}$ by any bijection. An active vertex then has the form $(x,(z,\varepsilon))$; give it label z. Let π denote this labeling map. Each label occurs exactly $b=2h$ times in each active set.

Define a t-partite graph Z on $A_1\cup\cdots\cup A_t$ by putting, for $u\in A_i$ and $w\in A_j$ with $i\neq j$,

$$uw\in E(Z)\quad\Longleftrightarrow\quad\pi(u)\pi(w)\in E(H).$$

The graph G retains all the edges of the modules and, between different active sets, takes the complement of Z. There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G.

We record three properties of Z. First, Z is K_t-free: the labeling map sends every clique injectively to a clique of H. Moreover, adding any missing edge between different parts of Z creates a K_t with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a K_{t-2} in $N_H(z) \cap N_H(z')$. Place its labels in the other $t-2$ parts. Together with u, w, they form the required clique in $Z + uw$. If the labels coincide, say both equal z, choose a vertex $z' \neq z$ not adjacent to z; this is possible because $d < v - 1$. The same common-neighborhood clique is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_{t-1} across any prescribed $t-1$ parts. Indeed, H has a missing edge; one endpoint together with a K_{t-2} in the common neighborhood gives a K_{t-1} in H. Its labels can be placed in any $t-1$ parts of Z.

Third, $D := \Delta(Z) = (t-1)bd = 2h(t-1)d$, so (3.1) gives $\ell > 2tD$.

The chromatic lower bound. Suppose that G has a $(2t+1)$-coloring. Fix a part A_i and any color α. Since its copy of S uses all $2t+1$ colors, some vertex x of S has color α. The ℓ active vertices in the row indexed by x avoid α, so the pigeonhole principle and $\ell > 2tD$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i, each used more than D times in A_i.

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the $2t$ chosen colors $\alpha_1, \beta_1, \ldots, \alpha_t, \beta_t$ are distinct, leaving a single color γ. The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 3(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_t in Z, a contradiction. Thus $\chi(G) \geq 2t+2$.

Colorings after edge deletion. Use the $(2t+1)$-color palette

$$\alpha_1, \beta_1, \ldots, \alpha_t, \beta_t, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G. By the first property of Z, there is a K_t in $Z + uw$ containing uw, with vertices $a_i \in A_i$. Apply Lemma 3(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i. The only color shared by different active sets is γ. Its t vertices are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained a $(2t+1)$-coloring of $G - e$.

Now let e lie within U_i. Apply Lemma 3(3) to color $U_i - e$, using only α_i, β_i on A_i. Choose a K_{t-1} in Z across the other $t-1$ parts. In each of those modules, apply Lemma 3(2) with the selected vertex as the only active vertex of color γ. Once again this is a proper $(2t+1)$-coloring of $G - e$.

These cases cover every edge of G. Recoloring one endpoint of a deleted edge with a new color gives $\chi(G) \leq 2t+2$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is $(2t+2)$-critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $a^2(1 - d/v)$ edges remain between each pair of active sets, proving (3.3). □

4 Proof of the main theorem

Proof of Theorem 1. First let $k = 2t + 2$ be even, with $t \geq 3$ fixed. By Alon, Erdős, Holzman, and Krivelevich [1, Theorem 7], for every sufficiently large v there is a K_t-saturated graph H_v on v vertices with maximum degree $d_v = O_t(\sqrt{v})$. Choose odd integers $h_v \sim v^{1/4}$. Then $h_v \to \infty$ and $h_v d_v / v = O_t(v^{-1/4}) \to 0$, so all hypotheses of Proposition 4 hold for sufficiently large v.

Apply the proposition and write $h = h_v$, $d = d_v$, $a = 2hv$, and $N = |V(G_v)|$. The order formula gives $N \sim ta$. Since $e(S) = O_t(h)$, $e(T) = O_t(v)$, and each module has $2a$ edges incident with its active set, the module edges number $O_t(a)$. Hence

$$\binom{t}{2} a^2 \left(1 - \frac{d}{v}\right) \leq e(G_v) \leq \binom{t}{2} a^2 + O_t(a).$$

As $d/v \to 0$, it follows that

$$\frac{e(G_v)}{N^2} \longrightarrow \frac{\binom{t}{2}}{t^2} = \frac{t-1}{2t} = \frac{k-4}{2(k-2)}.$$

For odd $k = 2t + 3 \geq 9$, join each graph in the even construction to K_1. The resulting graph is k-critical, has $N + 1$ vertices and $e(G_v) + N$ edges, and has the same limiting density, now equal to $(k - 5)/(2(k - 3))$. $\square$

Code availability

A Lean 4 formalization of Theorem 1 and Corollary 2, including a formal verification of the saturated-graph construction, is available at `https://github.com/FireflySentinel/erdos-917`.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the construction of critical graphs and the density estimates, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

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